Photosynthesis in winter

During the winter season, some kinds of plants retain their leaves—even though their photosynthesis slows down. Other species drop their leaves and become dormant, having stored up enough food to last them until spring.

Evergreen tree
Pine trees have tough needlelike leaves that can keep working even in freezing temperatures.

Deciduous tree
Many broad-leaved trees drop all their leaves at once in winter and grow a new set in spring.

Chemical reactions

Inside a chloroplast, a complex chain of chemical reactions takes place, which uses up water and carbon dioxide and generates sugar and oxygen. The light energy trapped by chlorophyll is first used to extract hydrogen from water, and expel the excess oxygen into the atmosphere. The hydrogen is then combined with carbon dioxide to make a kind of sugar called glucose. This provides the energy the plant needs for all the functions of life.